SEC : (P – 1 & P – 2) SR ELITE , AIIMS S60, MPL & LTC VAIDYAH **DATE : 09-04-2026**

NEET SPECIAL GRAND TEST – 07 KEY

BOT: 92, 94, 98, 99, 100, 102, 104, 106, 108, 109, 112, 114, 116, 117, 121, 123, 125, 127, 129, 132, 134, 136, 138, 141, 143, 145, 147, 149, 151, 153, 155, 157, 159, 161, 163, 165, 167, 169, 171, 173, 175, 177, 178, 179, 180

ZOO: 91, 93, 95, 96, 97, 101, 103, 105, 107, 110, 111, 113, 115, 118, 119, 120, 122, 124, 126, 128, 130, 131, 133, 135, 137, 139, 140, 142, 144, 146, 148, 150, 152, 154, 156, 158, 160, 162, 164, 166, 168, 170, 172, 174, 176

PHYSICS

1	2	2	2	3	1	3	3	1	4
4	4	2	1	4	1	3	3	2	2
1	3	3	2	2	4	1	1	4	2
2	4	1	3	1	3	3	4	4	1
4	2	1	4	2					

CHEMISTRY

1	2	4	3	2	2	1	1	1	4
1	4	1	4	4	2	3	3	3	4
3	1	1	3	1	3	1	4	3	1
1	1	4	3	4	3	3	4	1	1
4	3	1	3	2					

BIOLOGY

4	4	3	3	1	4	1	1	1	2
3	3	2	3	3	2	1	1	1	4
3	3	4	1	3	1	3	2	4	2
1	2	2	2	4	1	2	4	4	4
4	3	3	2	4	3	1	2	1	4
4	1	3	3	4	1	3	3	3	3
1	2	2	4	1	2	4	1	3	3
1	3	3	1	4	2	3	2	1	1
3	1	1	3	1	2	1	2	3	2

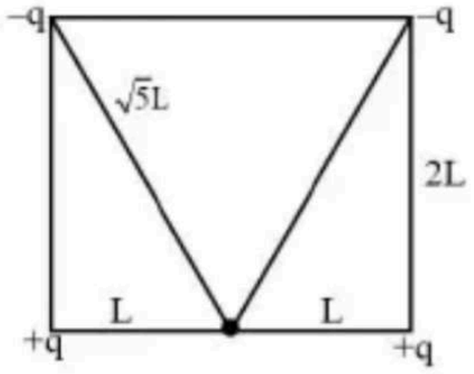
SOLUTIONS

PHYSICS

1. $\tan\theta_a = \tan\theta_m$

$$\frac{F_{air}}{mg} = \frac{F_{me}}{mg\left(1-\frac{d_t}{d_0}\right)}$$

2.



$$V = \frac{1}{4\pi\in_0}\frac{2q}{L}\left[1-\frac{1}{\sqrt{5}}\right]$$

3. C_1, C_2 & C_3 are in series

$$V_1 : V_2 : V_3 = \frac{1}{C} : \frac{1}{2C} : \frac{1}{3C}$$

$$\therefore V_2 = \frac{3V}{11}$$

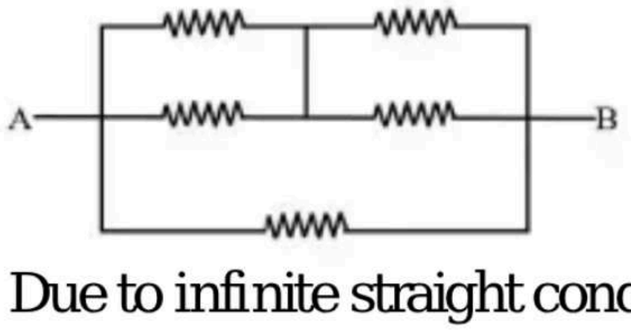
$$Q_2 = 2C\left(\frac{3V}{11}\right); Q_n = 4CV$$

4. To balance the bridge $\frac{6x}{6+x} = 2\Omega$

5. $i_1 = \frac{E}{R_1 + r}$

$$i_2 = \frac{E}{R_2} + r$$

6. Circuit can redrawn as



7. Due to infinite straight conductors

$\frac{2\mu_0 i}{4\pi R}\left(-\hat{k}\right)$; due to semicircle $\frac{\mu_0 i}{4R}\left(-\hat{i}\right)$

8. $v = \frac{mv}{qB}$

$mv = rqB$

$L = mvr$

9. $I = KH$

10. Given:

$I = 0.1\,\text{A}, \frac{dI}{dt} = 0.5\,\text{A/s}, P = 0.5\,\text{W}$

Power due to inductance:

$P = L \cdot I \cdot \frac{dI}{dt}$

Substitute values:

$0.5 = L \cdot 0.1 \cdot 0.5$

$L = \frac{0.5}{0.1 \times 0.5} = \frac{0.5}{0.05} = 10\,\text{H}$

Final Answer: 10 H

11. at resonance $i = \frac{V}{R}$; $V_R = \varepsilon_{rms}$

12. Induced emf ⇒ change in magnetic flux ($\Phi = BA\cos\theta$)

(A) Uniform motion in uniform field ⇒ Φ=constant ⇒ No emf

(B) Non-uniform speed ⇒ still Φ=constant (no change in B, A, θ) ⇒ No emf

(C) Rotating coil ⇒ θ changes ⇒ Φ changes ⇒ emf induced

(D) Changing area ⇒ A changes ⇒ Φ changes ⇒ emf induced

⇒ Correct options: C, D

13. Conceptual

14. Number of images:

$n = \frac{360^\circ}{\theta} - 1 \left(\text{since} \frac{360}{60} = 6 \text{ is even}\right)$

$\Rightarrow n = \frac{360}{60} - 1 = 6 - 1 = 5$

$\Rightarrow$ Number of images = 5

15. $\frac{1}{V}-\left(\frac{1}{-30}\right)=\frac{1}{10}$

$V = 15$ cm later screen moved by 45 cm, now $V = 60$ cm

$$\frac{1}{V}+\frac{1}{30}=\frac{1}{f_e}$$

$$\frac{1}{f_1}+\frac{1}{f_2}=\frac{1}{f_e}$$

16. Since $AQ = AR$

$r_1 = r_2 = r$

$$r=\frac{A}{2}$$

$$\mu=\frac{\sin i}{\sin r} \text{ and } 2i = A + D$$

17. $2I_0 = 4I_0\cos^2\frac{\phi}{2}$

$$\phi=\frac{\pi}{2}$$

$$\Delta x_1 = \phi\times\frac{\lambda}{2\pi} \text{ and } y_1=\frac{D}{d}\Delta x_1$$

Similarly find y_2

$$y_2 - y_1 = \frac{\alpha}{12}$$

18. Conceptual

19. de-Broglie wavelength,

$$\lambda=\frac{h}{\sqrt{2mE_3}}=\frac{hc}{E_p} \quad \text{or} \quad 2mF_e=\frac{E_p^2}{c^2}$$

But $E_e=\frac{1}{2}mv^2$ or $m=\frac{2E_e}{v^2}$

From eq. (i), we get

$$2\left[\frac{2E_e}{v^2}\right]E_e=\frac{E_p^2}{c^2} \text{ or } \frac{4E_e^2}{v^2}=\frac{E_p^2}{c^2}$$

or

$$\frac{E_e^2}{E_p^2}=\frac{v^2}{4c^2} \text{ or } \frac{E_e}{E_p}=\frac{v}{2c}$$

20. $P=\frac{E}{C}$ and $E=13.6\left[1-\frac{1}{16}\right]$ ev

21. NCERT Page no. 319

22. Due to forward biased during positive half cycle of input AC voltage, the resistance of p-n junction is low. The current in the circuit is maximum. In this situation, a maximum potential difference will appear across resistance connected in series of circuit. This result into zero output voltage across p-n junction.

Due to reverse biased during negative half cycle of AC voltage, the p-n junction is reverse biased. The resistance of p-n junction becomes high which will be more than resistance in series. That is why, there will be voltage across p-n junction with negative cycle in output.

23. $\overline{A.B}+\overline{C}$

$\bar{A}+\bar{B}+\bar{C}$

24. While forming depletion layer near the junction N-type becomes positive and P-type becomes negative.

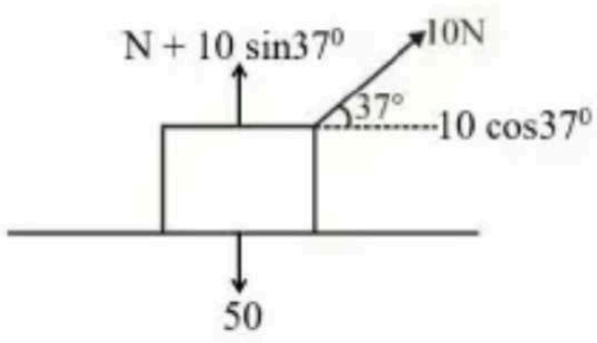
25. $40(g + a) = 480$

$10 + a = 12$

$a = 2$ m/s^2

$$a=\left(\frac{M-60}{M+60}\right)g$$

26.



$10\cos 37^{o} < f_{limiting}$

$\therefore$ friction $= 8$ N

27. $L.C = \frac{0.1}{10} = 0.01$ cm

1st reading $= 0.5 + 8(0.01) = 0.58$ cm

+ve correction $= 0.58 + 0.03 = 0.61$ cm

Similarly 2nd with correction $= 0.57$ cm

3rd with correction $= 0.59$ cm

mean $= 0.59$ cm

28. D.F of $P = ML^{-1}T^{-2}$

$$\frac{a}{L^6} = ML^{-1}T^{-2}$$

29. Conceptual

30. $mg - T = ma \rightarrow (1)$

$TR = I\alpha$

$$TR = mR^2\frac{a}{R}$$

$T = ma$ substitution in (1)

$$a = \frac{g}{2}$$

31. $V = 0 + a_1t_1$ for accolated motion

$0 = V - a_2t_2$ for retardation

$a_1t_1 = a_2t_2$

32. $\bar{v}_1 = 5i$ m/s

$\bar{v}_2 = 5j$ m/s

$$\bar{a} = \frac{\bar{v}_2 - \bar{v}_1}{t}$$

33. $\frac{1}{2}mu^2 = 90$

$u = 30$ m/s

$\frac{1}{2}mv^2 = 40$

$v = 20$ m/s

$v - u = at\ (t = 1)$

$30 - 20 = a(1)$

$a = 10$ m/s^2

$v^2 - u^2 = 2a\,s_1$

$(30)^2 - (20)^2 = 2(10)s_1$

$s_1 = 25$m

$(20)^2 - (0)^2 = 2a\,s_2$

$20 \times 20 = 2(10)\,s_2$

$s_2 = 20$

$s_1 + s_2 = 45$m

34. $V_\theta^2 = V_L^2 - 2gR[1 - \cos\theta]$

$\theta = 135^{o}$ with lower verticle

35. At centre of earth $g = 0$

36. To escape

$U + KE = 0 \qquad \left(\therefore KE = -\frac{1}{2}U\right)$

$KE = -U$

$KE = -(-mv^2) = mv^2$

37. $\frac{F}{\pi r^2} = 379 \times 10^6$

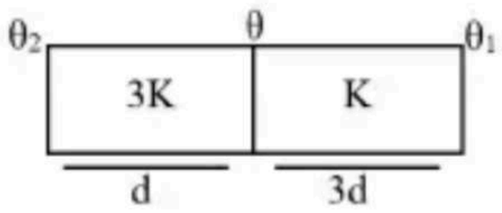
$\frac{7 \times 400}{22r^2} = 379 \times 10^6$

38. $V(0.9) = x(1) + (V - x)(0.8)$

39. $P \propto \frac{1}{r} \propto \frac{1}{V^{1/3}}$

40. $\propto = \frac{\Delta L_1 + \Delta L_2}{L_1 + L_2}$

41.



$\frac{\theta_2 - \theta}{\theta - \theta_1} = \frac{d}{3d} \times \frac{K}{3K}$

42. $dw = \frac{nR(T_1 - T_2)}{r-1} = \frac{nRT_1 - nRT_2}{r-1}$

$= \frac{P_1V_1 - P_2V_2}{r-1}$ $P_1 = 1.013$ $V_1 = 1$ lit

$P_2 = ?$ $V_2 = 3$ lit

$P_1V_1^r = P_2V_2^r$

43. $\frac{1}{2}m\omega^2\left(A^2 - y^2\right) = \frac{1}{2}m\omega^2y^2$

44. $\frac{V}{2L_0} = f$ half-length in side water

$$\frac{V}{4\frac{L_0}{2}} = \frac{V}{2L} = f^1$$

45. $V_r = \sqrt{\frac{3RT}{M}}$

$V_p = \sqrt{\frac{2RT}{M}}$

CHEMISTRY

46. $\lambda \propto n$

47. Molar mass of O_2 is 32g

48. $K^+, Ba^{+2}, F^-, Al^{+3}$

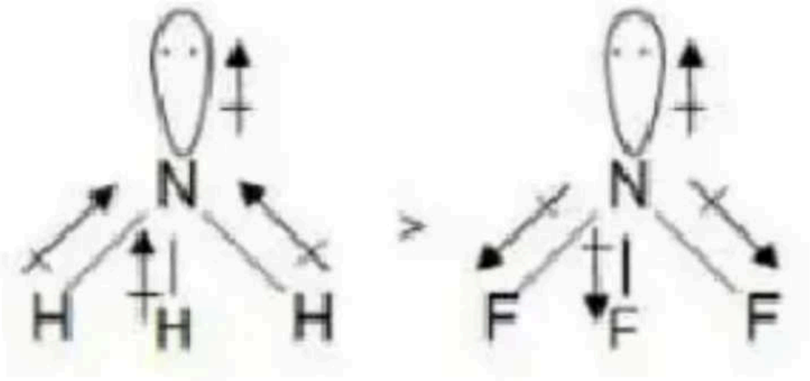
49. NH_3, CH_3^- are pyramidal shape (3 base pairs +1 lone pair) and contains 10 electrons each

50. On Pauling scale electronegativity of $F = 4.0, N = 3.0, H = 2.1$

Difference in electronegativity in N – F is greater in N – H bond

$\therefore$ N – F bond is more polar than N – H bond

Dipole moment



51. A) Only orbitals of same atom can intermix with each other to form hybrid orbitals

B) Any orbital whether half filled or vacant or completely filled can involve in hybridization

C) Each hybrid orbital can accommodate a maximum of two electrons with opposite sign

D) Hybrid orbitals will form sigma bond always (only in benzyne molecule a pi bond is formed due to $sp^2 - sp^2$ overlap)

52. Aufbau's principle is not violated

(1) In '1s' orbital both electrons have same spin. This violates Pauli's principle

(2) In '2p' subshell all the three unpaired electrons in degenerate orbitals must have parallel spin to have maximum exchange energy. This is Hund's rule. But one of the unpaired electron has opposite spin. This violates Hund's rule

53. $Ni^{2+} + DMG^- \longrightarrow \left[Ni(DMG)_2\right]\downarrow$

(Bright red precipitate)

54. Flame test :

Cu^{2+} is green flame with blue centre

Sr^{2+} is crimson red

Ca^{2+} is brick red

Ba^{2+} is apple green

55. $\Delta T_f = i \times K_f \times m$

$$0.2 = i \times 1.86 \times \frac{0.7}{93} \times \frac{1000}{42}$$

$$i = \frac{0.2 \times 93 \times 42}{1.86 \times 0.7 \times 1000}, \quad i = 0.6$$

$$i = 1 + \left(\frac{1}{n} - 1\right)\alpha$$

$$0.6 = 1 + \left(\frac{1}{2} - 1\right)\alpha$$

$\alpha = 0.8$

percentage of $\alpha = 80\%$

56. $t_{1/2} \propto a$

57. Water soluble vitamins are almost not stored in the body except vitamin B_{12}

58. $\Delta G^\circ = -RT\ln K$ .Using this one can calculate ΔG°

59. Keratin and myosin are fibrous proteins

Insulin and albumins are globular proteins

60. CH_3COONa is a salt of weak acid and strong base whose $P^H = 7 + \frac{1}{2}\left(P^{K_a} + \log C\right)$

61. $N_2O_4 \rightleftharpoons 2NO_2$

$t=0 \quad 1\,mol \quad 0\,mol$

$Kc = [B]^2 / [A]$

$Kc = (1.2)^2 / 0.4$

$Kc = 1.44 / 0.4$

$Kc = 3.6$

62. $\Delta H_{isomerisation} = (\Delta H_C)_R - (\Delta H_C)_P$

$-1359.2 - (-1336.0) = -23.2 kJ/mol$

63. $Cu^{+2} + 2e^- \longrightarrow Cu(s)$

On applying Nernst equation,

$E = E^o + \frac{0.0591}{2}\log\left(10^{-1}\right)$

$= +0.34 - 0.02955 = +0.31V$

64. $V_1 N_1 = V_2 N_2$

$1\left(10^{-6}\right) = 100(N_2)$

$N_2 = \left[H^+\right] = 10^{-8} M$

$\left[H^+\right]_{solution} = 10^{-8} + 10^{-7}$

$= 1.1 \times 10^{-7}$

$P^H = 7 = \log 1.1$

$= 6.96$

65. On comparing experimental results of I and II, we can conclude that Rate $\propto$ P (when [P] is made 4 times rate became 4 times)

On comparing experimental results of II and III, we can conclude that Rate $\propto Q^2$ (when [Q] is made 2 times rate became 4 times)

Based on above observations,

$Rate = K[P]^1[Q]^2$

66. Conceptual

67. $31g \longrightarrow -Q$

$31 \times 4g \longrightarrow ?$

$= -4Q$

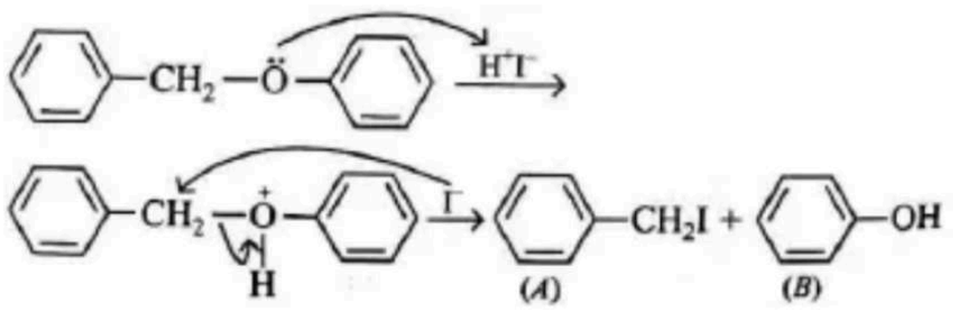
68. It is ideal solution

69. Percentage of $P = \frac{62}{222} \times \frac{0.22}{0.12} \times 100 = 51.20$

70. Conceptual

71. III is most stable due to complete octet

72. Alkyl aryl ethers are cleaved at the alkyl-oxygen bond due to more stable aryl-oxygen bond. The reaction yields phenol and alkyl halide.



73. Benzene $+Cl_2 \xrightarrow{AlCl_3}$ Chlorobenzene.

Benzene $+CH_3Cl \xrightarrow{AlCl_3}$ Toluene

Benzene $+ CH_3COCl \xrightarrow{AlCl3}$ Methyl phenyl ketone

Toluene $\xrightarrow[H^+]{KMnO_4/KOH}$ Benzoic acid

74. Conceptual

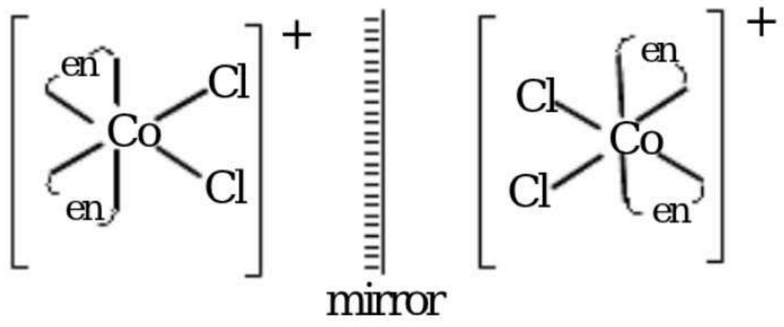
75. Conceptual

76. Ar is used in the laboratory for handling substances that are air sensitive

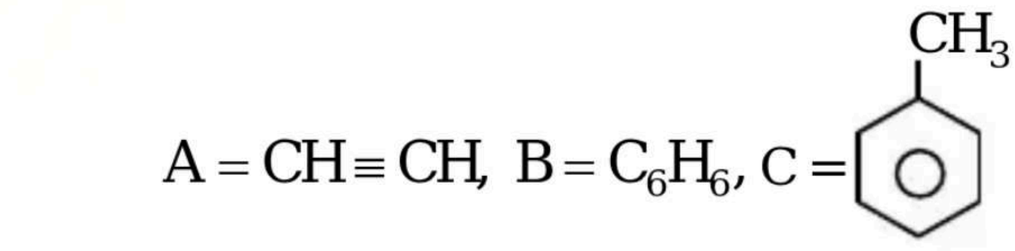
77. Conceptual

78.

a)



79.



$$A = CH \equiv CH,\ B = C_6H_6,\ C =$$

80. Conceptual

81.

$$CH_3COCl \xrightarrow[Pd\text{-}C/BaSO_4]{H_2} CH_3 - CHO - \text{Rosenmund reaction}$$

$$CH_3 - CHO \xrightarrow{\text{dil. NaOH}} CH_3 - \underset{OH}{CH} - CH_2 - CHO$$

$$\xrightarrow{\Delta} CH_3 - CH = CH - CHO$$

– Aldol conendensation

$$CH_3 - CH_2 - COOH \xrightarrow[RedP]{Br_2} CH_3 - \underset{Br}{CH} - COOH - \text{HVZ reaction}$$

$$CH_3 - CHO \xrightarrow[ConcHCl]{Zn-Hg} CH_3 - CH_3 - \text{ clemenson's reduction}$$

82. Conceptual

83. Conceptual

84. Ethane nitrile to ethanal - DIBAL-H

Allyl alcohol to propenal – PCC

But-2-ene to ethanal - O_3Zn/H_2O

85. Correct order is Ca < Al < C < O < F

86. Conceptual

87. Conceptual

88. $C_2O_4^{-2}$ is strong field ligand in Co^{+3} complexes

89. Conceptual

90. Stability of $(C_6H_5)_2\overset{+}{C}(CH_3) > (C_6H_5)\overset{+}{C}(CH_3)_2 > (C_6H_5)\overset{+}{C}H(CH_3)$

BIOLOGY

91. NCERT, Vol. II: Male reproductive system, 2nd & 3rd para

Rete testis connects seminiferous tubules with vasaefferentia

93. NCERT, Vol. II: Infertility, ART, 2nd para

ICSI is the suitable ART procedure to solve the problem of infertile due to tough corona radiata.

95. NCERT, Vol. II: Human health & disease, AIDS, 3rd para.

AIDS patients are more susceptible to various infections due to immune deficiency

96. NCERT, Vol. II: Reproductive health, IUDs.

The hormone releasing IUDs also promote phagocytosis of sperms but cannot prevent ovulation.

97. NCERT, Vol. II: Pregnancy & embryonic development, 4th para.

By the end of 12th week external genitalia are formed, so the sex of the foetus can be detected easily by sonography.

101. NCERT, Vol. I: Animal Kingdom & frog.

Frogs are poikilothermic, ureotelic tetrapods.

103. NCERT, Vol. I: Animal Kingdom

Spongilla - Fresh water sponge; *Meandrina*-Brain coral; *Adamsia*-Sea anemone; *Ctenoplana*-Sea walnut.

105. NCERT, Vol. I: Animal Kingdom

Open type of circulation is common for grasshopper, *Saccoglossus* and *Chaetopleura*.

107. NCERT, Vol. I: Animal Kingdom, Vertebrata

Electric organ- *Torpedo*; Pneumatic bones-*Neophron*; Without ear pinnae-*Ornithorhynchus*

108. The temperature at which pollen grains can be stored for several years in liquid nitrogen is -196°C. This process, known as cryopreservation, is used to preserve pollen grains and other biological materials at ultra-low temperatures to maintain their viability for extended periods.

Correct option is A

110. Varsity's Akash material: Tissues.

Cell junctions are highly specialised, structural modifications of the plasma membrane. They provide both structural and

functional links between individual cells of a tissue.

111. NCERT, Vol. I: Animal Kingdom, Symmetry.

In radial symmetry any plane passing through the central axis of the body divides the organism into two identical halves.

113. Varsity's Akash material: Tissues.

Intervertebral discs, eustachian tube and ear pinna are composed of cartilage

114. Explanation: In 1869, Friedrich Miescher discovered DNA and called it 'Nuclein,' marking the first identification of DNA.

115. NCERT, Vol. I: Body fluids & circulation, formed elements.

Basophils; Eosinophils; Monocytes; Lymphocytes; Neutrophils

117. Statement-II: Quaternary structure of a protein is not for biological activity. It may be made up of more than one polypeptides. Eg.: Haemoglobin. It may bring structural variations.

118. NCERT, Vol. II: Human health & disease, Common diseases in man.

Sleeping sickness and malaria are the vector-borne protozoan diseases.

119. NCERT, Vol. II: Human health & disease, Cancer.

HPV can cause cervical cancer. Cellular oncogenes are present in healthy cells. Metastasis is not the property of malignant tumors. MRI is used to detect physiological changes in organs or tissues.

120. NCERT, Vol. II: Active & Passive immunity

Antitetanus serum, antibodies that cross placenta and colostrum are the examples for passive immunity

122. NCERT, Vol. II: Human health & disease, Common diseases in man.

Ascariasis-Internal bleeding, fever, blockage of intestinal passage

Filariasis-Inflammation of lymphatic vessels

Dengue-Joint pains, pain behind eyes

Ringworm-Dry, scaly lesions on skin, nail, scalp etc.

123. Statement B asserts that only 80S ribosomes are composed of two subunits. This is incorrect because both 80S ribosomes (found in eukaryotes) and 70S ribosomes (found in prokaryotes, mitochondria, and chloroplasts) are composed of two subunits. 80S ribosomes consist of a 60S and a 40S subunit, while 70S ribosomes consist of a 50S and a 30S subunit.

124. NCERT, Vol. II: Population interactions, Predation.

Predators reduce the intensity of competition among competing prey species. Predators are prudent.

126. NCERT, Vol. II: Ecosystem, 2nd para.

The solar input, the cycle of temperature, photoperiod and day length regulate the rate of function of the pond ecosystem.

128. NCERT, Vol. II: Ecosystem, Structure & function, 2nd para.

All ecosystems are self-sustainable entities, they don't depend on outside resources, except for light. In any ecosystem, complex interactions exist due to flow of energy and materials.

130. NCERT, Vol. II: Ecosystem, 2nd para.

The following four basic components of the ecosystem are seen to function as a unit.

Productivity = Input; Decomposition = Output; Energy flow =Relationships established within the system; Cycles, chains and webs =Energy flow:

131. NCERT, Vol. II: Biodiversity, Species-area relationship.

Relationship between species richness (s) and area (A) for a wide variety of taxa turns out to be a straight line on a logarithmic scale.

133. NCERT, Vol. II: Biodiversity conservation, 2nd para.

Pollination comes under broadly utilitarian services of an ecosystem

135. NCERT, Vol. II: Biodiversity conservation, How do we conserve biodiversity.

A Biodiversity hotspot may include one or two national parks in it. A large national park may include one or two sanctuaries in it. Eg. Gir National Park includes Sasan Gir sanctuary.

As the Biodiversity knows no political boundaries, its conservation is the responsibility of all nations.

All the 34 Biodiversity hot spots put together cover less than 2% of the Earth's land area.

137. Assertion is true. Drones are the male bees in a honeybee colony. They are haploid, having a single set of chromosomes, which they inherit from only one parent—their mother (the queen bee). Drones develop from eggs laid by the queen that are not fertilised by sperm, hence they have no father.

The Reason accurately explains the Assertion. Parthenogenesis is a form of asexual reproduction in which an egg develops into an organism without fertilisation. In honeybees, parthenogenesis leads to the development of drones, which inherit genetic material exclusively from their mother, the queen.

139. NCERT, Vol. II: Menstrual cycle.

GnRH and oestrogen levels increase during follicular phase of menstrual cycle.

140. NCERT, Vol. II: Mendelian disorders, 4th para.

Sickel-cell anaemia is an autosomal recessive disorder

141. Breakdown of proton gradient is possible while they move across the membrane from lumen to stroma due to concentration gradient.

142. NCERT, Vol. II: DNA finger printing.

If a germ cell mutation does not seriously impairs reproduction, it can be inherited by offspring from the parents

143. RuBisCo, can't act as oxygenase in B.S cell of C_4 plants as there is high CO_2 con., where 'RuBisCo' shows greater affinity to high CO_2 con. Thus photorespiration is suppressed.

144. NCERT, Vol. II: Biotechnology and its Applications.

Alpha lactalbumin - Balanced diet;

Genetically engineered Insulin – Avoids allergic reactions;

Alpha-interferons –To treat Cancer

Transgenic animals – To study Alzheimer's disease

146. NCERT, Vol. II: Biotechnology and its Applications. Molecular diagnostics

ELISA test is used to detect antigens or antibodies

147. Mobile carrier for transport of electrons between complex III and IV of mitochondrial ETS is Cyt c.It is located towards the outer surface of the inner membrane of mitochondria

148. NCERT, Vol. II: Factors affecting H&W equilibrium.

Sometimes the allelic frequencies in a population are changed due to some chance events without the involvement of natural selection. It is called the genetic drift. Founder effect is a kind of genetic drift.

150. NCERT, Vol. II: Evolution, Adaptive radiation.

Wings of butterfly and wings of bird are analogous structures. They differ in their structure but similar in their function

Development of similarities among unrelated species due to same selection pressure is called convergent evolution

152. NCERT, Vol. II: Evolution, H&W principle, 5th para, Brief account of evolution.

Homo erectus was the first to eat meat. Adaptive ability is heritable.

154. Statement I is incorrect. When reptiles came down, mammals took over this earth.

Statement II is correct. Small-sized reptiles of the Mesozoic Era still exist today.

156. NCERT, Vol. I: Breathing & exchange of gases, Transport of gases.

The condition that increases O_2 binding to hemoglobin at the lung surface is high pH ie., low acidity.

158. NCERT, Vol. I: Mechanism of Breathing.

When diaphragm contracts, the volume of the thoracic chamber increases in the antero-posterior axis.

160. NCERT, Vol. I: Body fluids & circulation, ECG

Standard ECG does not reflect atrial repolarisation as this deflection is overshadowed by QRS complex of waves.

162. NCERT, Vol. I: Body fluids & circulation, Cardiac cycle.

In a cardiac cycle end of atrial systole is the end of ventricular diastole. End of ventricular systole is the beginning of joint diastole. End of joint diastole is the end of atrial diastole and is the beginning of atrial systole.

164. NCERT, Vol. I: Excretory products and their elimination, 2nd para.

Hemidactylus is a reptile which excretes uric acid.

166. NCERT, Vol. I: Skeletal system, Axial skeletal system.

The first 7 pairs of ribs in man are double headed (bicephalic) and attached to sternum on ventral side (vertebrosternal type).

168. NCERT, Vol. I: Mechanism of muscle contraction, 2nd para.

White muscle fibres are fast twitch muscles with high intensity of contraction and anaerobic muscles. So they quickly undergo fatigue. A skeletal muscle contains both red muscle fibres and white muscle fibres in different proportions

169. There is no involvement of 'RNA primers' in the transcription of mRNA in eukarotes.

170. NCERT, Vol. I: Generation and Conduction of nerve impulse, 2nd para.

The rise in the stimulus-induced permeability to Na^+ is extremely short lived as it is followed by a rise in permeability to K^+ to outside.

172. NCERT, Vol. I: Chemical coordination, Pituitary gland.

LH stimulates the secretion of androgens in males whereas in females it promotes ovulation, formation and maintenance of corpus luteum. FSH and androgens regulate spermatogenesis.

174. NCERT, Vol. I: Chemical coordination.

Androgens, cortisol and erythropoietin all are involved in erythropoiesis.

176. NCERT, Vol. II: Organisms& Populations, Life history variations.

Life history traits of organisms have evolved in relation to the constraints imposed by the abiotic and biotic components of the habitat. This is to achieve a high r value, the Darwinian fitness

177. Chromogenic substance is used to select transforments from non transforments (recombinants)